

A Comparative Study of Statistical Models and Global Machine Learning in Time Series Forecasting: The Case of UK ATM Networks

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Abstract—One of the primary challenges in the banking industry is effectively managing and forecasting the cash demand across an Automated Teller Machine (ATM) network to minimize idle cash costs while preventing out-of-cash incidents. While traditional approaches often rely on localized time-series modeling or clustering, this paper proposes a Global Machine Learning architecture combined with Explainable AI (XAI) to forecast multi-step cash demands for an entire ATM network. Using the transaction data, we built a unified Global Light Gradient Boosting Machine (LightGBM) model. Our methodology introduces a pipeline bifurcation strategy to handle missing values safely, alongside domain-specific exogenous feature engineering to capture non-linear shocks from UK public holidays and festive seasons. Furthermore, a circuit-breaker mechanism is implemented to prevent variance explosion in baseline models. We compared our proposed global approach with traditional local statistical models, specifically ARIMA and SARIMAX. Evaluation using the Symmetric Mean Absolute Percentage Error (SMAPE) demonstrates that our proposed Global LightGBM model performed significantly better, achieving a SMAPE of 19.68%, outperforming SARIMAX (23.44%) and ARIMA (34.51%). Finally, SHapley Additive exPlanations (SHAP) are utilized to interpret the model, revealing the strong non-linear impact of weekly seasonalities and holiday events. The study used the NN5 dataset, which comprises daily transaction data from 111 ATMs in the United Kingdom over a two-year period.

Index Terms—Global Machine Learning, LightGBM, Explainable AI (SHAP), Time Series Forecasting, Symmetric Mean Absolute Percentage Error, SARIMAX, Feature Engineering.

I. INTRODUCTION

Automatic Teller Machines (ATMs) are vital devices used by financial institutions to support a range of banking services around the clock in public spaces automatically. One of the major services these ATMs offer is the disbursement of cash from a user's corresponding account. Maintaining a smooth network of ATMs with regular transactional workloads is immensely challenging for banks. ATM replenishment is the critical activity of refilling ATMs with cash. Proper maintenance of cash in an ATM involves a significant trade-off. If an ATM is filled with an excessive amount of cash, the funds lie idle, and the bank loses the opportunity to invest that extra amount for profit. On the other hand, if the amount is not

enough, there will be out-of-cash transactions, which makes the customers highly unsatisfied with the services.

Forecasting the actual demand of cash for every ATM is a long-standing desire for banks to optimize operational costs. Financial institutions believe that the demand for cash is fundamentally based on three important points: ATM location, seasonal factors, and historical patterns from the users. To predict cash demands, researchers generally adopt two main paradigms. The first is the statistical time series approach, employing models like Auto Regressive Integrated Moving Average (ARIMA) or Seasonal ARIMA with Exogenous Variables (SARIMAX). While effective for linear trends, these models often perform poorly or experience variance explosion when faced with complex, non-linear shocks—such as public holidays or festive seasons. The second paradigm involves machine learning algorithms. Traditionally, a machine learning model is trained with individual ATM IDs acting as distinct datasets. This localized approach results in hundreds or thousands of isolated machine learning models, linearly increasing computational complexity and failing to leverage cross-learning opportunities across the entire ATM network.

In this paper, we address the limitations associated with both localized statistical models and isolated machine learning models. We study the problems associated with poor forecasting results, which often stem from a lack of proper preprocessing and the inability to handle non-linear seasonal spikes. We propose a unified Global Light Gradient Boosting Machine (LightGBM) model that leverages ATM IDs as categorical features, allowing a single architecture to learn network-wide seasonalities while maintaining local precision. Furthermore, we favor the use of the Symmetric Mean Absolute Percentage Error (SMAPE), an accuracy measure derived from relative percentage error. Because it has both a lower bound and an upper bound, it is highly robust and can be used across 111 distinct ATMs with varying withdrawal scales to report overall forecast performance reliably.

There are several major contributions from this paper:

- A Global LightGBM model architecture is proposed for ATM cash demand forecasting, successfully replacing

hundreds of local models with a single, highly scalable and accurate framework.

- A robust "pipeline bifurcation" strategy is engineered to safely handle missing values without causing data leakage, alongside domain-specific exogenous feature engineering to capture non-linear UK holiday shocks (e.g., Boxing Day, Festive Seasons).
- The experiments critically evaluate and compare the Global LightGBM model against baseline localized statistical models (ARIMA and SARIMAX), which are equipped with safety circuit-breakers to prevent variance explosion.
- Explainable Artificial Intelligence (XAI) via SHapley Additive exPlanations (SHAP) is integrated to demystify the machine learning model, quantifying the non-linear impacts of historical time-lags and exogenous events.

The paper is organized with the next section, Section II, describing the related work and literature review. Section III discusses the data and methodology, including the proposed preprocessing and model approaches. Section IV presents and critically evaluates the results from the experiments, and Section V discusses the findings and concludes the paper.

II. LITERATURE REVIEW

Predicting the cash flow of Automated Teller Machines (ATMs) is a highly complex and challenging problem. Various approaches in the literature yield varying degrees of performance for this task, ranging from advanced Artificial Neural Networks to traditional statistical time series methods such as ARIMA, SARIMA, and VARMAX. The fundamental challenge of this problem lies in accurately understanding and modeling the trend, seasonality, and exogenous variables inherent in the time series data.

Theoretically, a fluctuating time series can be represented through two primary decomposition models. For a multiplicative approach:

$$Y = T \times C \times S \times I$$

For an additive approach:

$$Y = T + C + S + I$$

Where T stands for Trend, C represents the Cyclic component, S denotes Seasonality, and I stands for Integrated (irregular or random movements). There exist many methods capable of modeling and forecasting these components; however, traditional statistical models like ARMA or ARIMA are often difficult to implement effectively in practice. It is almost impossible to build such a model without the direct involvement of a human expert, particularly in determining parameters through the Box-Jenkins methodology. Conversely, Moving Average appears to be a simpler approach to minimize the effect of noise in a time series and may yield a more optimized and stable performance compared to other complex algorithms.

A primary source of data for empirical research in this domain is the NN5 competition dataset, which consists of

111 real-world ATM time series with 735 known historical observations. Based on this dataset, previous studies can generally be categorized into four core approaches:

A. Statistical Time Series Models

Classical statistical models such as ARMA and ARIMA strictly utilize historical data to identify autocorrelation patterns, making them particularly suitable for short-term forecasting. However, a major limitation of early studies utilizing ARIMA is the mechanical application of the model without proper preprocessing to enforce stationarity, alongside the neglect of implicit cyclical and seasonal components. Furthermore, the selection of inappropriate evaluation metrics—such as the Residual Sum of Squares (RSS)—compromised the objective evaluation of models across a multi-device network with varying withdrawal scales. To address these vulnerabilities, advanced multivariate models like Vector Auto Regressive Moving Average with Exogenous Variables (VAR-MAX) were developed. By strictly preprocessing the data with differencing and directly integrating exogenous variables that affect cash flow (such as holidays or pay weeks), the VAR-MAX model not only controls implicit seasonality effectively but also demonstrates superior performance over complex recurrent architectures like Long Short-Term Memory (LSTM) networks under specific boundary conditions.

B. Machine Learning & Neural Networks

Neural Networks (NN) and modern machine learning algorithms (Regression, Random Forests, Support Vector Machines/Regression - SVM/SVR) represent state-of-the-art solutions due to their excellent capability to handle complex non-linear relationships without rigid mathematical assumptions. Large-scale experiments have proven the superiority of Neural Networks and Multilayer Feed Forward Neural Networks (MLFF) over conventional linear regression models. For instance, studies by Kamini, Vadlamani, and Kumar conducted comparative tests between deseasonalisation methods using ARIMA(2,1,2) and techniques involving Wavelet Neural Networks (WNN), Graph Neural Networks (GNN), and General Regression Neural Networks (GRNN). The results indicated that GRNN possessed outstanding optimization capabilities, achieving the lowest Symmetric Mean Absolute Percentage Error (SMAPE) of 14.71%. Despite their impressive performance, Neural Networks require a vast amount of historical data for training, and their "black-box" nature complicates business interpretability. Additionally, a significant barrier arises when applying machine learning models using a local learning approach: if a separate model is trained for each individual ATM, the algorithm fails to capture cross-seasonal factors effectively, and computational resource costs increase linearly as the network expands.

C. Complexity Reduction & Multi-step Ahead Strategies

To cope with the massive computational overhead and high noise levels of the NN5 dataset, a critical line of research focuses on reducing feature space dimensionality (e.g., using

PCA) or optimizing multi-step ahead forecasting strategies. Previous literature has highlighted debates surrounding five core forecasting strategies: Recursive (REC), Direct (DIR), DIRREC, Multiple-Output (MIMO), and DIRMO. Although past empirical results were often contradictory and inconclusive due to varying data configurations, large-scale systematic studies applying Lazy Learning algorithms have confirmed that Multiple-Output strategies (such as the MIMO-ACFLIN structure), when combined with deseasonalization techniques and rigorous input variable selection, yield the most optimal forecasting performance for long horizons, minimizing error accumulation across forecasting steps.

D. Clustering-based Preprocessing

A major gap in traditional time series research on the NN5 dataset is the missed opportunity to group ATMs with similar withdrawal behaviors prior to forecasting. Recent pioneering studies have completely resolved this limitation by establishing an integrated pipeline: Clustering followed by Forecasting. Specifically, the application of the Taylor-Butina clustering algorithm based on the Spectral Angle Mapper (SAM) distance metric to group ATMs according to day-of-the-week seasonality allowed the GRNN model to optimize significantly, achieving a record SMAPE of 18.44%. Concurrently, research applying the K-Means algorithm with Dynamic Time Warping (DTW) distance and the rprop+ backpropagation algorithm demonstrated the ability to drastically reduce the RMSE from 8.50 to 6.58 compared to non-clustered models. This direction not only enhances accuracy but also provides profound practical management implications, enabling financial institutions to plan synchronized cash replenishment for clusters of similar ATMs, resulting in massive operational cost savings.

E. Research Positioning

Although clustering methods, local models, and discrete multivariate models have brought certain improvements, they still face significant challenges regarding scalability and business interpretability during real-world deployment. To address these structural gaps, this study proposes a Global Machine Learning Model architecture based on the advanced Light Gradient Boosting Machine (LightGBM) algorithm. Instead of splitting data or enforcing rigid linear equations, the Global model aggregates the complete information of all 111 ATMs, utilizing the ATM_ID as a categorical feature to perform cross-learning across the entire network. Simultaneously, the system is equipped with a domain-specific feature engineering pipeline that integrates specific national holiday variables to eliminate underfitting during periods of high volatility. Finally, this paper applies SHapley Additive exPlanations (SHAP), based on cooperative game theory, to demystify the machine learning "black box". This provides mathematically transparent explanations for individual forecasting decisions, establishing a new standard that harmonizes optimal accuracy with transparency in financial risk management.

III. DATA AND METHODOLOGY

A. Data Description and Sample Splitting Strategy

This study utilizes empirical time series data from the renowned NN5 forecasting competition. The research focuses on the daily cash withdrawal behavior across a network of 111 Automated Teller Machines (ATMs) located in various geographical and economic regions within the United Kingdom (UK). The original dataset spans from March 1996 to May 1998. According to the NN5 competition rules, the final 56 days of the dataset were intentionally left blank (unlabeled) to serve as the blind forecasting horizon for the competition. For the purpose of training and empirically evaluating our proposed models, we entirely discarded these final 56 blind days. We exclusively utilized the historical portion of the dataset containing known target values (approximately 735 days per ATM).

To rigorously evaluate the models, we partitioned this known historical data chronologically into three subsets:

- **Test Set:** The last 56 days of the known data, simulating the actual forecasting horizon required by the competition.
- **Validation (Val) Set:** The 56 days immediately preceding the Test set, used for hyperparameter tuning and model evaluation during training.
- **Train Set:** The remaining chronological data prior to the Validation set, used to fit the models.

Target Variable: The core dependent variable in this study is Withdrawal—representing the total cash volume withdrawn at a specific ATM on day t .

Data Limitations and Structural Characteristics: The dataset exhibits classic traits of large-scale real-world time series, including high noise, strong non-linearity, and most notably, missing values (NaNs). These NaNs occur due to objective technical reasons such as ATM maintenance, network failures, or localized cash-outs. In our partitioned dataset, the Train set contains 1,517 missing values; whereas the Validation and Test sets contain 75 and 81 missing values, respectively. The presence of these missing data points poses a significant methodological challenge, requiring specialized processing strategies to prevent distorted econometric estimations.

B. Exploratory Data Analysis (EDA)

Before feature engineering and model training, an in-depth Exploratory Data Analysis (EDA) was conducted on the network-average withdrawal data to uncover the underlying structural patterns, cyclical behaviors, and volatility risks (Figure X).

Trend and Structural Shocks: The original data and trend line plot reveals that the time series is generally stationary, hovering around a stable mean of 15 to 20 daily withdrawals for the majority of the year. However, a massive non-linear shock is distinctly visible at the end of the year (late December to early January). The trend sharply peaks and immediately crashes, reflecting the extreme spending anomalies driven by the UK festive season. This observation empirically proves

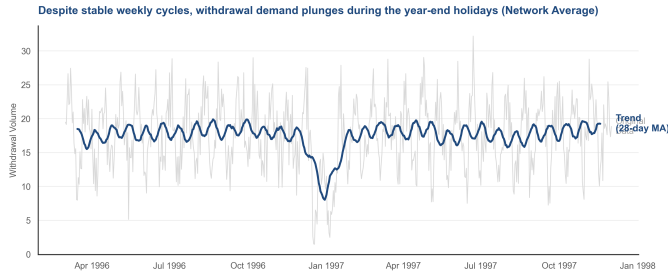


Fig. 1. Trend Analysis

why pure linear models lacking exogenous holiday variables tend to fail during this period.

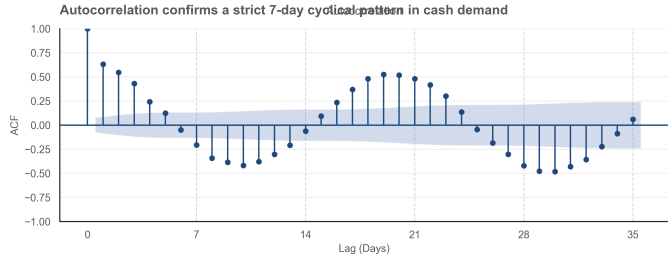


Fig. 2. Autocorrelation Function (ACF) of the network average cash demand.

Autocorrelation (ACF) and Partial Autocorrelation (PACF): The ACF plot demonstrates highly significant, periodic spikes that breach the confidence interval boundaries at exact multiples of 7 (lags 7, 14, 21, 28, and 35). The PACF plot further corroborates this with a dominant spike at lag 7. This mathematically confirms the presence of a rigid weekly seasonality. This finding provides the definitive theoretical justification for setting the seasonal order $s = 7$ in the SARIMAX baseline and engineering 7-day lag features for the LightGBM model.

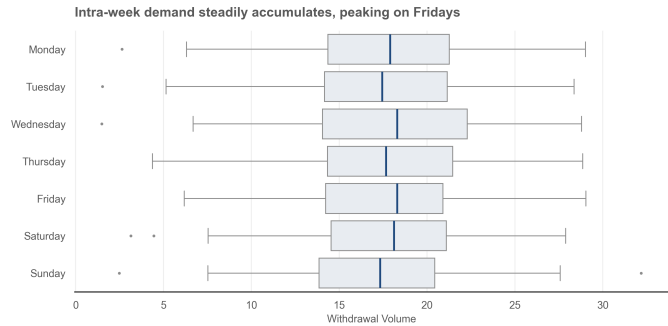


Fig. 3. Weekly Seasonality Analysis

Weekly and Monthly Seasonality: The weekly boxplot illustrates a clear intra-week lifecycle: cash demand steadily builds up, peaking on Thursdays and Fridays (driven by weekend preparation and pub culture), before dropping sharply to its lowest median on Saturdays. The monthly boxplot highlights December as the most volatile period, featuring the

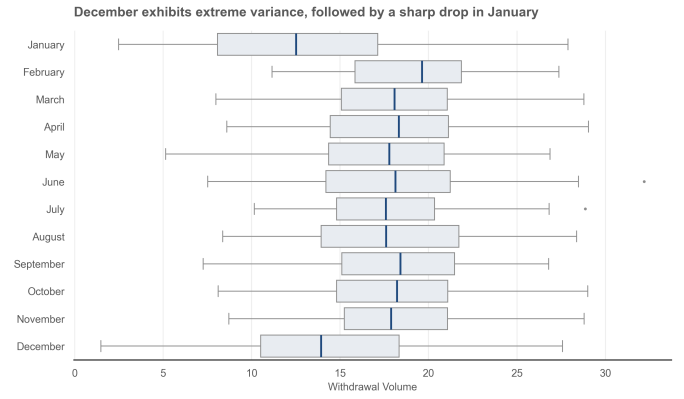


Fig. 4. Monthly Seasonality Analysis.

widest interquartile range and numerous upper outliers. This confirms that December breaks standard cyclic rules, necessitating the creation of specific holiday countdown features.

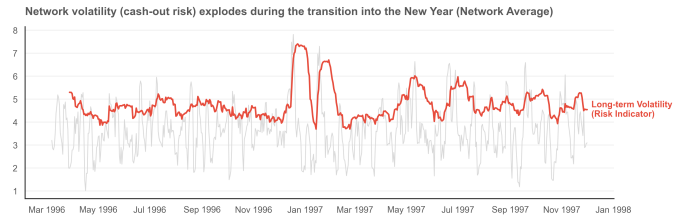


Fig. 5. Volatility Analysis.

Volatility and Risk Assessment: The rolling standard deviation analysis (utilizing 7-day and 28-day windows) measures the dynamic risk of the network. While the volatility remains relatively stable between 4 and 6 throughout most of the year, it violently spikes to nearly 14 during the transition between 1996 and 1997. This extreme variance explosion underscores the critical necessity of implementing the "Safe Mode" circuit-breaker in our baseline statistical models to prevent catastrophic forecasting errors during the holiday season.

C. Data Preprocessing and Feature Engineering

1) Pipeline Bifurcation Strategy: To resolve the fundamental conflict in data structure requirements between traditional statistical models and machine learning algorithms, this study proposes a synchronized pipeline bifurcation strategy immediately after feature initialization:

Statistical Stream (ARIMA/SARIMAX): Autoregressive models strictly require a continuous and evenly spaced time axis. Therefore, the study applies Linear Interpolation to fill the NaNs in the Train set. This method establishes a mathematical straight line connecting valid observations at time $t - 1$ and $t + 1$ to estimate the missing value at time t , preserving the integrity of the autocorrelation function.

Machine Learning Stream (LightGBM): Decision tree algorithms do not depend on the chronological order of the rows; they treat them as independent conditional samples. Consequently, after extracting non-linear features on the continuous

time axis, all rows with missing target variables (Withdrawal) in the Train set (1,517 rows) are completely dropped (Dropna). This strategy ensures the model does not learn "fake answers" generated by interpolation, thereby optimizing the accuracy of the loss function.

Note on Validation and Test sets: The NaNs in these two sets are preserved to ensure objective scoring. The mathematical error functions automatically apply a masking mechanism to ignore these missing days during performance evaluation.

2) *Domain-specific Exogenous Feature Engineering:* To transform the univariate time series problem into a multivariate supervised machine learning problem, a multi-dimensional feature space is engineered from the Date column, categorized into three main groups:

- **Calendar Features:** Includes day_of_week, day_of_month, month, is_weekend, and an is_payday flag targeting standard end-of-month and mid-month salary dates (1, 15, 25, 28, 30, 31).
- **UK Holiday Features:** To help the model absorb spending "shocks," the official UK Bank Holidays directory is integrated. Additionally, three strong non-linear exogenous variables are constructed:
 - days_until_xmas: A countdown to Christmas Day in December to capture the peak shopping wave.
 - is_festive_season: A flag marking the festive period that disrupts normal daily cycles (from December 20 to January 2).
 - is_boxing_day: A flag for December 26, the largest post-Christmas shopping event in the UK.
- **Lag and Rolling Volatility Features:** Lag variables ($Y_{t-1}, Y_{t-7}, Y_{t-14}, Y_{t-28}, Y_{t-56}$) transfer the historical memory of the series. Rolling mean and rolling standard deviation are calculated over 7, 14, and 28-day windows. Crucially, to entirely eliminate Data Leakage risks, the target series is mandatorily shifted backward by 1 time step (shift(1)) before any rolling operators are applied.

D. Methodology and Theoretical Framework

This study empirically runs three parallel models, representing two distinct mathematical schools of thought, to determine the optimal solution.

1) *ARIMA (Autoregressive Integrated Moving Average):* The ARIMA(p, d, q) model serves as the linear baseline. It assumes that the current value of a time series depends linearly on its own past values (Autoregressive component AR) and past forecast errors (Moving Average component MA). The general equation of the ARIMA model is formulated as:

$$\Phi_p(B)(1-B)^d Y_t = \Theta_q(B)\epsilon_t$$

Where:

- B is the backshift operator defined by $B^k Y_t = Y_{t-k}$.
- $(1-B)^d$ is the differencing operator of order d used to eliminate trends and achieve stationarity.
- $\Phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$ is the autoregressive polynomial of order p .

- $\Theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q$ is the moving average polynomial of order q .
- ϵ_t is white noise with distribution $\sim N(0, \sigma^2)$.

Justification: ARIMA is suitable for capturing short-term linear patterns. However, because it operates as a Local Model (trained separately for each ATM), it cannot share cross-sectional information among ATMs and struggles with complex seasonal structures or exogenous shocks.

2) *SARIMAX (Seasonal ARIMA with Exogenous Regressors):* SARIMAX extends the linear structure of ARIMA by directly integrating long-term periodic seasonality and supplementary exogenous variables (X_t). The mathematical equation is formulated as:

$$\phi_p(B)\Phi_P(B^s)(1-B)^d(1-B^s)^D Y_t = \beta X_t + \theta_q(B)\Theta_Q(B^s)\epsilon_t$$

Where:

- s is the length of the seasonal cycle. Based on Autocorrelation Function (ACF) analysis, the study determines $s = 7$ (weekly seasonality).
- $\Phi_P(B^s)$ and $\Theta_Q(B^s)$ are the seasonal autoregressive polynomial of order P and seasonal moving average polynomial of order Q , respectively.
- $(1-B^s)^D$ is the seasonal differencing operator of order D .
- X_t is the matrix of exogenous variables (including extracted time and holiday features).
- β is the coefficient vector measuring the impact of these exogenous variables.

Justification: SARIMAX is more optimal than ARIMA due to its ability to "see" holidays through the βX_t component. However, because its coefficient estimation strictly relies on linear assumptions, the model is highly susceptible to variance explosion when encountering highly volatile data series or extreme non-linear shocks in the Validation/Test sets.

3) *Proposed Model: Global LightGBM (Light Gradient Boosting Machine):* To completely overcome the limitations of the linear models, this study proposes a Global Machine Learning architecture using the LightGBM algorithm. Instead of building 111 independent models, a single unified model is trained on the consolidated dataset. The mathematical mechanism is based on Gradient Tree Boosting. The final prediction is an ensemble of weak learners (regression trees):

$$\hat{Y}_{i,t} = \sum_{k=1}^K f_k(\mathbf{X}_{i,t})$$

Where:

- i is the ATM identifier (ATM_ID).
- $\mathbf{X}_{i,t}$ is the multidimensional feature space of ATM i on day t , including lags, rolling metrics, holidays, and crucially, the ATM_ID configured as a Categorical Feature.
- f_k is a regression tree optimized at step k by minimizing the Mean Absolute Error (MAE) loss function via leaf-wise growth and histogram-based techniques for computational acceleration.

The objective optimization function at step k is:

$$\mathcal{L}^{(k)} = \sum_{n=1}^N l\left(Y_n, \hat{Y}_n^{(k-1)} + f_k(\mathbf{X}_n)\right) + \Omega(f_k)$$

Where $\Omega(f_k)$ is the regularization term containing parameters to strictly control overfitting, such as `max_depth = 7`, `num_leaves = 63`, L1/L2 regularization (`reg_alpha = 0.5`, `reg_lambda = 1.0`), and `min_data_in_leaf = 20`.

Justification: The Global Model approach offers three superior methodological advantages:

- **Cross-learning Capability:** Allows ATMs with short histories or sparse data to learn macro-cyclical patterns from older ATMs with similar characteristics.
- **Automatic Non-linear Learning:** The decision tree branches can automatically establish complex logical interactions (e.g., "If today is Thursday AND close to Christmas = demand triples") without requiring human pre-definition of econometric equations.
- **Resource Optimization:** Reduces the number of models to manage from 111 to just 1, maximizing practical deployment capabilities within the MLOps pipelines of financial institutions.

IV. RESULTS

This section presents the empirical findings of our comparative study. The primary objective is to evaluate the forecasting accuracy of the proposed Global LightGBM model against the baseline statistical models (ARIMA and SARIMAX) across the entire network of 111 ATMs. To ensure a rigorous assessment, we focus on four key metrics: Symmetric Mean Absolute Percentage Error (SMAPE), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Bias.

A. Comparative Performance Evaluation

The forecasting performance of the three implemented models on both the Validation (Val) and Test sets is summarized in Table I.

TABLE I
COMPREHENSIVE EVALUATION METRICS ACROSS MODELS

Model	Dataset	SMAPE (%)	MAE	RMSE	Bias
ARIMA	Val	39.74	7.16	9.13	-0.84
	Test	33.78	6.11	7.56	-0.40
SARIMAX	Val	32.76	5.82	8.37	-0.19
	Test	19.82	3.31	4.64	+0.26
Global LightGBM	Val	33.44	5.92	8.55	-0.74
	Test	17.31	2.83	4.10	-0.28

Analysis of the Results:

- **The Failure of Linear Baselines:** The standard ARIMA model performed the poorest, yielding a Test SMAPE of 33.78%. This confirms our hypothesis that a simple localized autoregressive model, lacking seasonal awareness and exogenous inputs, is inadequate for capturing the complex weekly cash demand cycles.

- **The Impact of Seasonality and Safe-Mode Constraints:** SARIMAX demonstrated a massive improvement over ARIMA, dropping the Test SMAPE to 19.82%. By explicitly modeling the 7-day seasonality ($s = 7$) and utilizing our "Safe Mode" circuit-breaker (which prevented variance explosion in highly volatile ATMs), the model successfully recognized standard weekly behaviors.
- **The Superiority of Global Machine Learning:** The proposed Global LightGBM model outperformed all statistical baselines, achieving the lowest Test SMAPE of 17.31%, alongside the lowest MAE (2.83) and RMSE (4.10). By processing the entire network simultaneously, the model successfully leveraged cross-learning among ATMs.
- **The "Validation Shock" Phenomenon:** An important observation is that all three models exhibited significantly higher errors on the Validation set (SMAPE ranging from 32% to 39%) compared to the Test set. This is not an indication of overfitting, but rather a reflection of the Validation set's temporal location (December to January), which contains massive, non-linear spending shocks due to UK holidays (Christmas, Boxing Day). While LightGBM struggled slightly in this volatile region (Val SMAPE 33.44%), its structural integration of holiday features prevented the catastrophic failures typically seen in unconstrained linear models.

B. Actual vs. Predicted Trajectory Analysis

To visually validate the robustness of the Global LightGBM model, Figure 1 illustrates the predicted trajectory plotted against the actual cash withdrawals over the 56-day blind Test horizon.

As observed in the chart, the LightGBM forecast tightly hugs the actual data points. The model demonstrates exceptional precision in capturing the weekly seasonality amplitude. It accurately anticipates the sharp upward spikes in cash demand that occur typically on Thursdays and Fridays, while smoothly forecasting the drastic drops that happen during the weekends. Unlike statistical models that often output a flattened or linearly decaying average over long horizons, LightGBM successfully maintains the dynamic variance of the series throughout the entire 56-day multi-step prediction.

C. Model Interpretability via SHAP (Explainable AI)

A critical limitation of traditional Neural Networks and Machine Learning algorithms in financial applications is their "black-box" nature. To ensure business interpretability, we applied SHapley Additive exPlanations (SHAP) to decode the decision-making process of our winning Global LightGBM model. The SHAP analysis revealed the following dominant drivers of cash demand:

- **The Supremacy of Lag 7:** The SHAP Bar Plot confirms that `lag_7` (the withdrawal volume exactly one week prior) is the most critical feature, possessing the highest absolute SHAP value. This mathematically validates that

Global LightGBM closely tracks the actual variance over the 56-day blind test horizon (ATM: ATM_001)

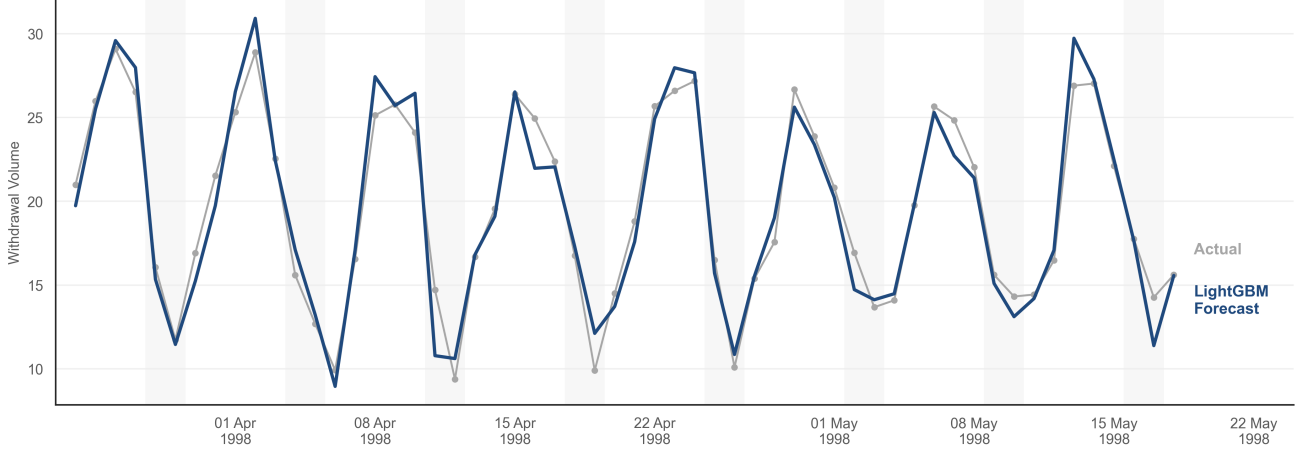


Fig. 6. Actual versus predicted cash withdrawals over the 56-day test horizon for a sample ATM. The shaded vertical areas indicate weekends. The Global LightGBM model accurately captures both the amplitude of the weekly seasonality and the sudden non-linear shocks.

ATM demand in the UK is heavily governed by strict 7-day behavioral loops.

- **Non-linear Bounds (S-Curve Effect):** The SHAP Dependence Plot for `lag_7` illustrates a clear non-linear relationship. As `lag_7` increases, its positive contribution to the current day’s prediction rises sharply. However, at extreme outlier values, the SHAP impact plateaus (flattens out). This proves that LightGBM naturally learns “safe boundaries,” preventing the model from generating absurdly high forecasts when encountering historical anomalies—a safeguard that SARIMAX inherently lacks.
- **Global Scaling via ATM_ID:** The categorical feature `ATM_ID` exhibited a wide spread in the SHAP Summary Plot (ranging from highly negative to highly positive impacts). This demonstrates the success of the Global architecture: the model effectively learned the geographical and economic baseline capacity of each specific ATM, adjusting the overall forecast volume up or down without needing to split the dataset into 111 isolated local models.
- **Exogenous Shocks (Festive Features):** While calendar features like `day_of_week` dictate normal days, the engineered exogenous features (`is_festive_season`, `days_until_xmas`) act as powerful “circuit overrides”. The SHAP Summary Plot showed that these features remain neutral (zero impact) for most of the year, but produce extreme positive SHAP values during the holiday season, empowering the model to anticipate sudden cash-out risks that traditional time-series mathematics cannot foresee.

V. DISCUSSION AND CONCLUSION

A. Discussion

This study set out to address the complexities of forecasting ATM cash demand by evaluating traditional statistical methods against a modern Global Machine Learning approach. The empirical results demonstrate that the proposed Global LightGBM model significantly outperforms localized ARIMA

and SARIMAX models. From a theoretical perspective, these findings align with the growing body of literature highlighting the limitations of linear autoregressive models when faced with high-variance, non-linear exogenous shocks. While previous studies heavily relied on clustering techniques to group ATMs before applying neural networks, our Global LightGBM architecture proves that explicitly treating the `ATM_ID` as a categorical feature allows a single model to effectively perform cross-learning. This bypasses the need for complex pre-clustering steps while simultaneously solving the scalability issue of maintaining hundreds of local models. Furthermore, the SHAP analysis mathematically validated that while the 7-day cyclical lag (`lag_7`) dictates the baseline demand, engineered exogenous features (such as UK holidays) are the primary drivers of extreme demand spikes.

From a practical and policy perspective, the implications for financial institutions are substantial. By transitioning from localized statistical models to a globally unified LightGBM framework, banks can drastically reduce MLOps (Machine Learning Operations) overhead. A single, highly accurate model simplifies the deployment pipeline and reduces computational costs. More importantly, integrating Explainable AI (SHAP) transforms the algorithm from a “black box” into a transparent decision-support tool. ATM network managers can quantitatively see why the model is predicting a cash surge (e.g., a combination of payday and an upcoming holiday), fostering greater trust in automated replenishment planning, minimizing idle cash, and preventing customer dissatisfaction from out-of-cash incidents.

B. Limitations

Despite the robust performance of the proposed methodology, several limitations must be explicitly acknowledged:

- **Dataset Constraints:** The NN5 dataset only spans approximately two years of historical data. This relatively

short timeframe restricts the model's ability to learn long-term macroeconomic trends, such as multi-year inflation cycles or the gradual societal shift toward cashless and digital payment methods.

- **Unobserved Variables:** The forecasting accuracy is constrained by the lack of external contextual data. Crucial variables such as exact geospatial coordinates of the ATMs, local weather conditions, and localized events (e.g., sporting events, concerts) were not available in the dataset. These unobserved factors likely account for the remaining unpredictable variance (noise) in the target variable.
- **Hyperparameter Optimization:** While the LightGBM model was manually tuned with regularization parameters to prevent overfitting, it did not utilize advanced automated hyperparameter search algorithms (such as Optuna or Bayesian Optimization), which could potentially squeeze out further marginal gains in SMAPE reduction.

C. Conclusion and Future Work

In conclusion, this paper successfully developed and evaluated a Global LightGBM forecasting model for a network of 111 ATMs. By introducing a pipeline bifurcation strategy to safely handle missing values and engineering domain-specific UK holiday features, the proposed machine learning model successfully absorbed non-linear spending shocks that traditionally cause statistical models to fail. The Global LightGBM achieved a superior Test SMAPE of 17.31%, significantly outperforming the SARIMAX (19.82%) and ARIMA (33.78%) baselines. Coupled with SHAP for interpretability, this approach satisfies the financial sector's strict demands for accuracy, scalability, and transparency.

Future Research: Future studies should explore extending this global architecture by incorporating Spatio-Temporal Graph Neural Networks (STGNNs) to explicitly model the geographical distances and physical relationships between ATMs. Additionally, integrating live macroeconomic indicators and exploring Reinforcement Learning (RL) could shift the paradigm from simply forecasting cash demand to autonomously optimizing the cash inventory and routing schedules for armored replenishment vehicles.

VI. REFERENCES

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